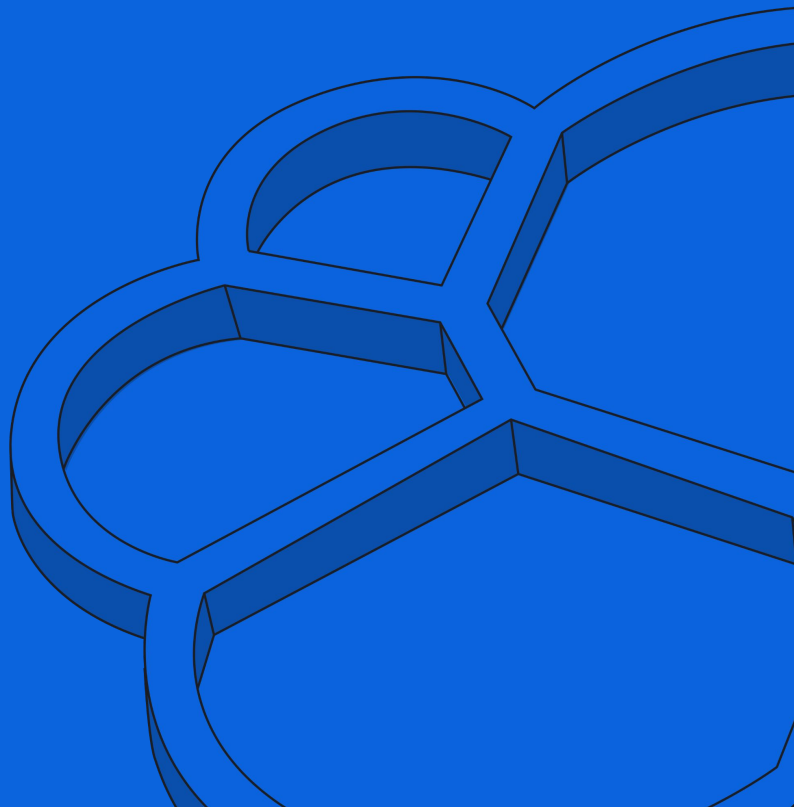
What's New in Elastic Search

Version 9.3

Roberto Arico - Senior Solutions Architect



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Workshop Agenda

Simplify

ES|QL Readable
Queries

LOOKUP JOIN
One query

From multiple API calls to single-query efficiency

Optimize

Workflows
Automation

Inference Endpoint

From complex external tools to native automation

Innovate with AI

Agent Builder
Ask in English

ELSER
Search by meaning

From keyword matching to semantic understanding

Today's Core Philosophy

Search **finds** the insight.

Workflows **acts** on it.

Agent Builder **explains** it.

Simplify:
ES|QL + Workflows

Optimize:
Automated Response

Innovate with AI:
Natural Language Investigation

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Today’s Format

4 sections with *2 parts*

Presentation **5 min**

Presentation & Live Demo.
Understanding the ‘Why’ and the ‘How’.

Workshop **10 min**

Hands-on Challenges. Accessing the lab Environment and running the code.

The Challenge: Review Bomb

"Review Bombing" is a coordinated attack where bad actors flood a business with fake negative reviews to manipulate reputation

Restaurants:
Yelp, Google, TripAdviser

E-commerce:
Amazon, Walmart, Home Depot

Apps:
App Store, Google Play

B2B:
Salesforce, Microsoft

Erodes consumer trust one fake star at a time

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The High Cost of Fake Reviews

Review integrity is not just a data annoyance. It is a financial imperative. When trust erodes, revenue follows. Bad actors use coordinated attacks to manipulate perception across platforms from Amazon, to Google to the App Store

5-9%

Revenue loss per 1-star drop

Platform Churn

Users abandon platforms when they encounter fake narratives.

Scale Challenge

Millions of reviews removed manually. Manual triage cannot keep pace with automated attacks.

Workshop Journey

1. Detection Logic

Write ES|QL queries to identify patterns.

Correlates Review Content with User Trust Scores

2. Workflow Def

Define automation logic

Scheduled execution every 5 minutes orchestrates response

3. AI Investigation

Build an AI Agent with custom tools

Natural Language analysis using custom AI tools

4. Simulation

Launch the live attack lifecycle

End-to-End Response

Know Your Data

Using Yelp Dataset (Philadelphia Subset)

Scope:

- 100 Businesses
- ~76K Users
- ~149k Reviews

Indices:

- businesses, users, reviews, incidents, notifications

Trust Score Logic:

- Low trust (<0.3): New/Inactive accounts
- High trust (>0.7): Established contributors

Defend this beloved business!!

Reading Terminal Market

(4.5 Stars)

5,700+ Reviews

Active

Food Court | Landmark | Philadelphia

Let’s Get Hands-On

Time to Simplify, Optimize and Innovate with AI

1. Log into your workshop environment
   a. **https://ela.st/za-community-mar2026**
   b. available for 48 hours
2. Mission:
   a. Detect a Review Bomb
   b. Contain it automatically
   c. Generate an incident report

Search *finds*. Workflows *acts*. Agent Builder *explains*

Challenge 1: Know your data

1. Detection Logic

Write ES|QL + LOOKUP JOIN to identify patterns.

Correlates Review Content with User Trust Scores

Activities:

- Getting to Know your data
  - Background
  - Data Model
- Explore it
- Understand Trust scores
- Review Patterns
- Semantic Search
- Enrichment with LOOKUP JOIN

Simplify Detection with ES|QL

Find the attack without complex syntax

Traditional queries can be verbose. ES|QL pipes data logically, making attack hunting intuitive

```
FROM reviews
| WHERE rating == 1
| STATS count() BY user_id
| SORT count() DESC
| LIMIT 10
```

Run

Workshop Action: Identify clusters of users posting identical reviews in seconds.

Trigger

Action

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The Power of LOOKUP JOIN

Correlate reviews with user reputation in a single query

Legacy approach (Application-side Join)

JetBrains Mono

```
1. GET /reviews?time=now-30m
2. Extract user_ids list
3. GET /users?ids=[list]
4. For each review: match user_trust_score
5. If score < 0.3: flag_review()
```

High Latency & Complexity

ES|QL Approach Elastic 9.3

JetBrains Mono

```
FROM reviews
| WHERE @timestamp > NOW() - 30m
| LOOKUP users ON user_id
| WHERE user_trust_score < 0.3
| STATS attack_count = COUNT(*) BY business_id
```

✅ Single Query. Millisecond Response. Zero Data Movement.

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Challenge 2: Workflows

2. Workflow Def

Define automation logic

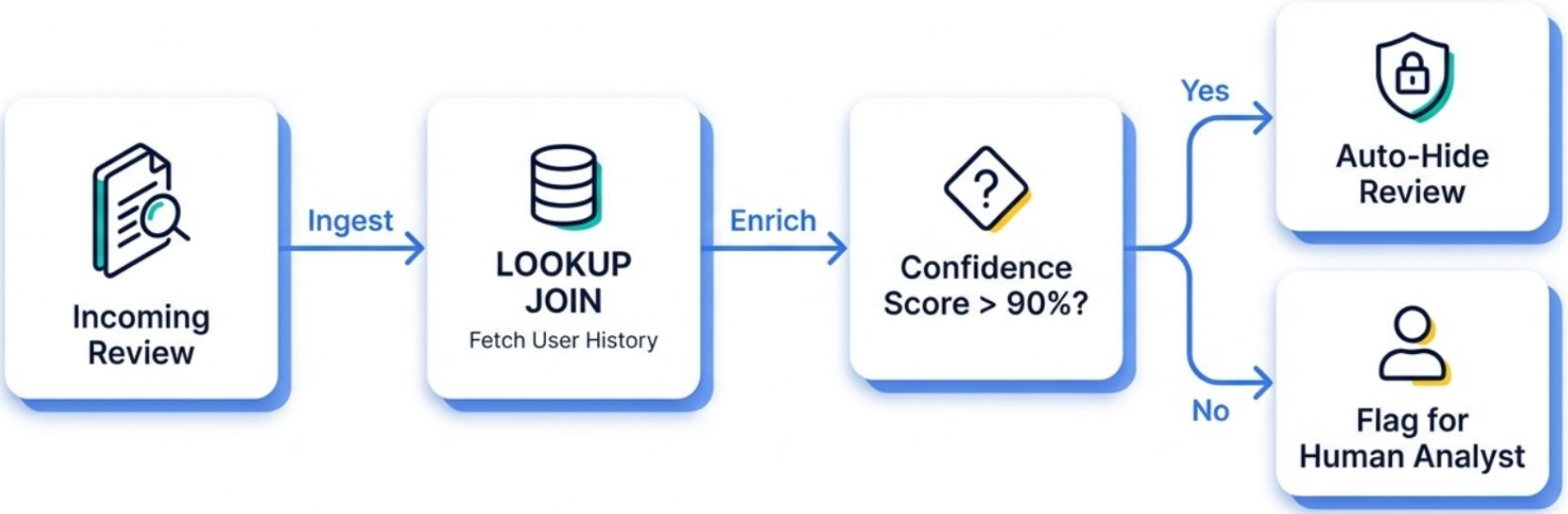
Scheduled execution every 5 minutes orchestrates response

Activities:

- Create an Automated workflow that detect negative review campaign and responds in real-time
- Understand the Workflow Structure

Optimize response with Workflows

Automate the takedown and enrich the context.



Workshop action: Build a workflow to hold attack reviews & create incident

Automated Response

Defining the logic of a review bomb

The Trigger

<15 Low-Trust 1-star reviews submitted in <1 min.

Workflow Action

1. Reviews marked ‘Held’
2. Businesses marked “PROTECTED”
3. Rating average frozen at 4.5 stars.

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Challenge 3: Agent Builder

3. AI Investigation

Build an AI Agent with custom tools

Natural Language analysis using custom AI tools

Activities:

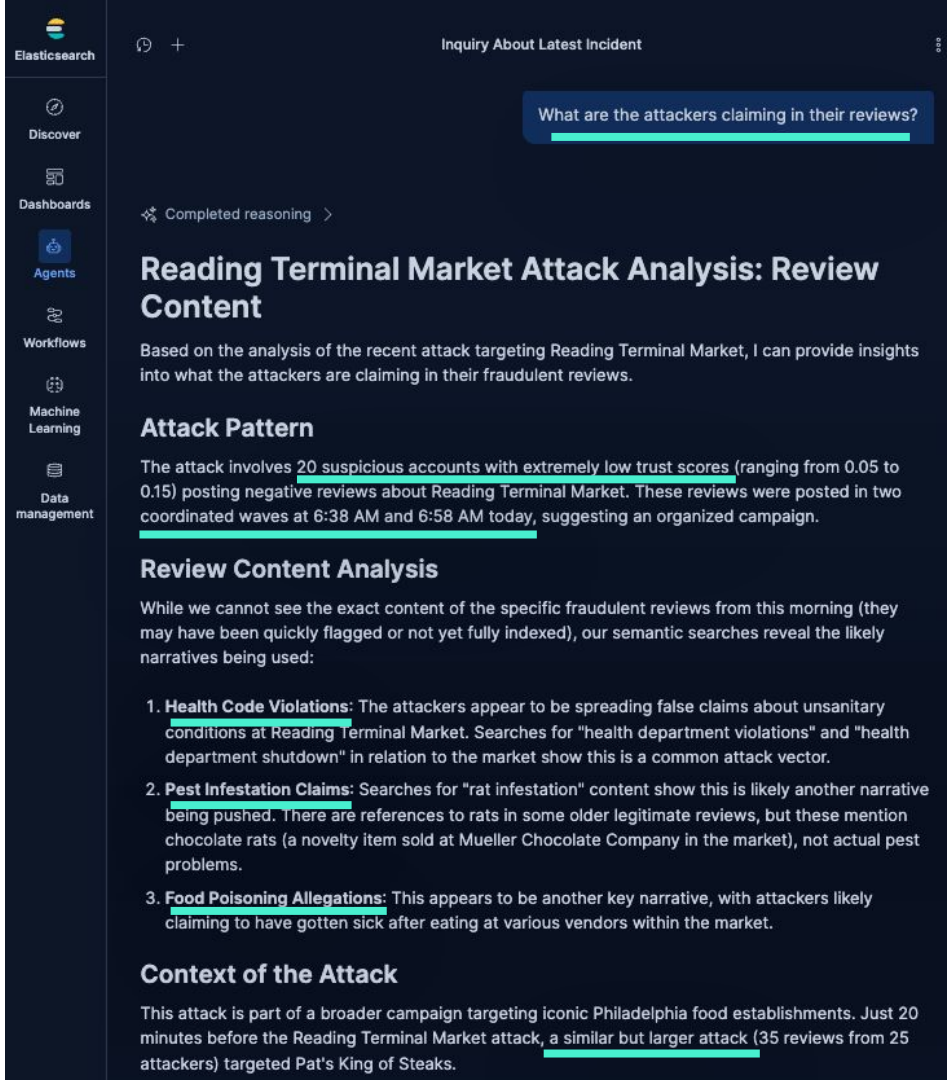
- Background
- Create Tools
- Create Agent

Investigates negative review campaigns against businesses. Can summarize incidents, analyze attacker patterns, and find similar malicious reviews

Innovate: Investigate with Agent Builder

Ask
"Who?"
"What?"
"Where?"
"When?"
"Why?"
"How?"
in plain English

Workshop Action: Deploy an AI Agent grounded in your private logs to generate incident reports automatically



Challenge 4: E2E Scenario

4. Simulation

Launch the live attack lifecycle

End-to-End Response

Activities:

- View ElasticEats
- Check baseline stats
- Launch an attack
- Watch the Workflow detection
- Investigate with your custom AI Agent
- Resolve the incident

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Thank *you!*

Labs

Available for next 48 hours

Code

`https://github.com/leungsteve/elastic-workflow-workshop`

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